

YEDITEPE UNIVERSITY

FACULTY OF ENGINEERING

DEPARTMENT OF INDUSTRIAL ENGINEERING

DECISION ANALYSIS OF INVESTING IN A NATURAL GAS PLANT VERSUS A SOLAR PV FARM

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1. PROBLEM STATEMENT

GreenWave Energy Inc. faces a strategic imperative to expand generation capacity by 2028 to meet a projected 20 % growth in regional peak demand, and must make a mutually exclusive choice between two project alternatives: building a 500 MW combined-cycle natural gas plant (required capital $450 million) or developing a 300 MW utility-scale solar photovoltaics (PV) farm (required capital $350 million).

If the gas plant is selected, CFO Helena Wang and her team must first decide on financing: either issuing ten-year green bonds with a coupon rate of 3.8 % (20 % likelihood) or 4.2 % (80 % likelihood), implying a total interest expense of $171 million or $189 million respectively; or obtaining a ten-year bank loan at a fixed interest rate of 6.5 %, which implies a total interest expense of $185 million. Once financing is secured, volatile gas markets drive the plant's gross inflows: $700 million if gas averages $3/MMBtu (20 % likelihood), $760 million if gas averages $5/MMBtu (50 % likelihood), and $820 million if gas averages $7/MMBtu (30 % likelihood).

If the solar PV farm is chosen, the investment—financed entirely by equity—would generate $425 million if electricity averages $45/MWh (30 % likelihood), $485 million if electricity averages $55/MWh (40 % likelihood), or $545 million if electricity averages $65/MWh (30 % likelihood).

2. DECISION TREE AND POSSIBLE STRATEGIES

2.1. INFLUENCE DIAGRAM

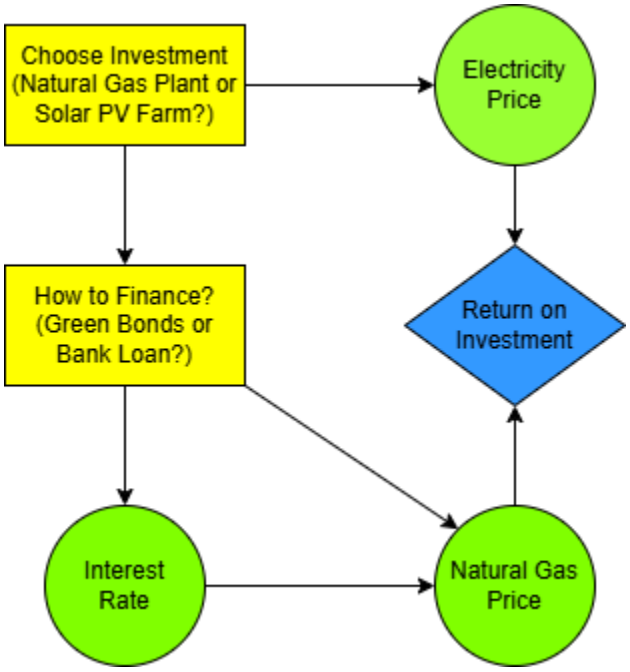


Figure 2.1. Influence diagram of the problem

2.2. DECISION TREE

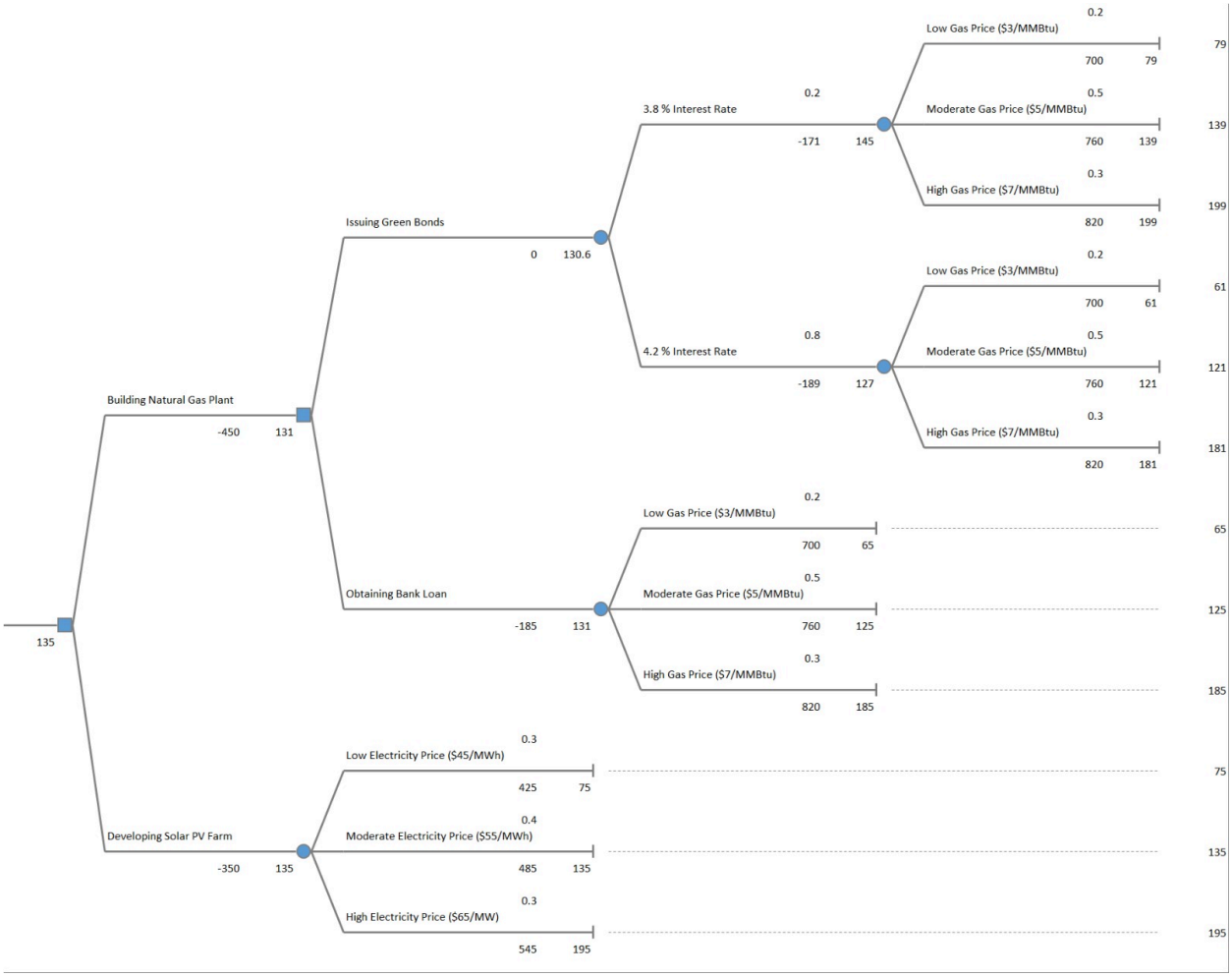


Figure 2.2. Decision tree of the problem

GreenWave Energy must choose one of the following mutually exclusive strategies to meet future demand:

S1: Build the Gas Plant + Issue Green Bonds

S2: Build the Gas Plant + Obtain Bank Loan

S3: Develop the Solar PV Farm

The decision tree begins with a decision node where the company selects one of the three strategies. The tree then branches based on uncertain market conditions and financing terms, leading to different financial outcomes. Each branch has:

- Uncertainties associated with market scenarios
- Net payoffs (gross inflows minus investment and financing costs)
- EMVs, which guide the optimal decision

2.2.1. Strategy 1: Build Gas Plant + Issue Green Bonds

Bond Interest Rate Uncertainty:

- 3.8% → $171 M interest expense (20%)
- 4.2% → $189 M interest expense (80%)

Gas Price Uncertainty:

- $3/MMBtu → $700 M inflow (20%)
- $5/MMBtu → $760 M inflow (50%)
- $7/MMBtu → $820 M inflow (30%)

For 3.8% interest (20%):

- Net Payoffs:
    - $700 M – $450 M – $171 M = $79 M
    - $760 M – $450 M – $171 M = $139 M
    - $820 M – $450 M – $171 M = $199 M
- EMV (at 3.8%): (0.2×$79 M) + (0.5×$139 M) + (0.3×$199 M) = $15.8 M + $69.5 M + $59.7 M = $145.0 M

For 4.2% interest (80%):

- Net Payoffs:
    - $700 M – $450 M – $189 M = $61 M
    - $760 M – $450 M – $189 M = $121 M
    - $820 M – $450 M – $189 M = $181 M
- EMV (at 4.2%): (0.2×$61 M) + (0.5×$121 M) + (0.3×$181 M) = $12.2 M + $60.5 M + $54.3 M = $127.0 M

EMV (S1) = (0.2 × $145.0 M) + (0.8 × $127.0 M) = $29.0 M + $101.6 M = $130.6 M

2.2.2. Strategy 2: Build Gas Plant + Obtain Bank Loan

Gas Price Uncertainty:

- $3/MMBtu → $700 M inflow (20%)
- $5/MMBtu → $760 M inflow (50%)
- $7/MMBtu → $820 M inflow (30%)

Net Payoffs:

- $700 M – $450 M – $185 M = $65 M
- $760 M – $450 M – $185 M = $125 M
- $820 M – $450 M – $185 M = $185 M

EMV (S2) = (0.2×$65 M) + (0.5×$125 M) + (0.3×$185 M) = $131 M

2.2.3. Strategy 3: Develop Solar PV Farm

Electricity Price Uncertainty:

- $45/MWh → $425 M inflow (30%)
- $55/MWh → $485 M inflow (40%)
- $65/MWh → $545 M inflow (30%)

Net Payoffs:

- $425 M – $350 M = $75 M
- $485 M – $350 M = $135 M
- $545 M – $350 M = $195 M

EMV (S3) = (0.3×$75 M) + (0.4×$135 M) + (0.3×$195 M) = $135.0 M

EMVs of Possible Strategies:

1. EMV (S1) = $130.6 M
2. EMV (S2) = $131 M
3. EMV (S3) = $135.0 M

The Solar PV Farm is the best strategy, with the highest EMV of $135 million. However, the Bank Loan option becomes more competitive, with an updated EMV of $131 million—now closer to the solar option than before. The Green Bond remains the least favorable.

The optimal strategy is “S3: Develop the Solar PV Farm”.

2.3. RISK PROFILES AND DOMINANCE

Expected monetary value (EMV) rankings tell us which project offers the highest average payoff, but they say nothing about how those pay-offs are distributed. A strategy delivering $135 M on average may still be unattractive if it occasionally drops deeply into losses, whereas a lower-EMV alternative might appeal to a risk-averse board if its worst-case outcome is far less severe.

To capture this dimension, we construct cumulative distribution functions (CDFs) for the net profits of all three strategies. Each CDF traces, for every profit level, the probability that actual results will be no greater than that level—effectively visualising downside risk and upside potential in a single curve. We then apply the concept of dominance.

2.3.1. Strategy 1: Build Gas Plant + Issue Green Bonds

Net Profit ($M)	Probability
61	0.16
79	0.04
121	0.40
139	0.10
181	0.24
199	0.06

Net Profit ($M)	Cumulative Probability
61	0.16
79	0.2
121	0.6
139	0.7
181	0.94
199	1

2.3.2. Strategy 2: Build Gas Plant + Obtain Bank Loan

Net Profit ($M)	Probability	Cumulative Probability
65	0.20	0.20
125	0.50	0.70
185	0.30	1.00

2.3.3. Strategy 3: Develop Solar PV Farm

Net Profit ($M)	Probability	Cumulative Probability
75	0.30	0.30
135	0.40	0.70
195	0.30	1.00

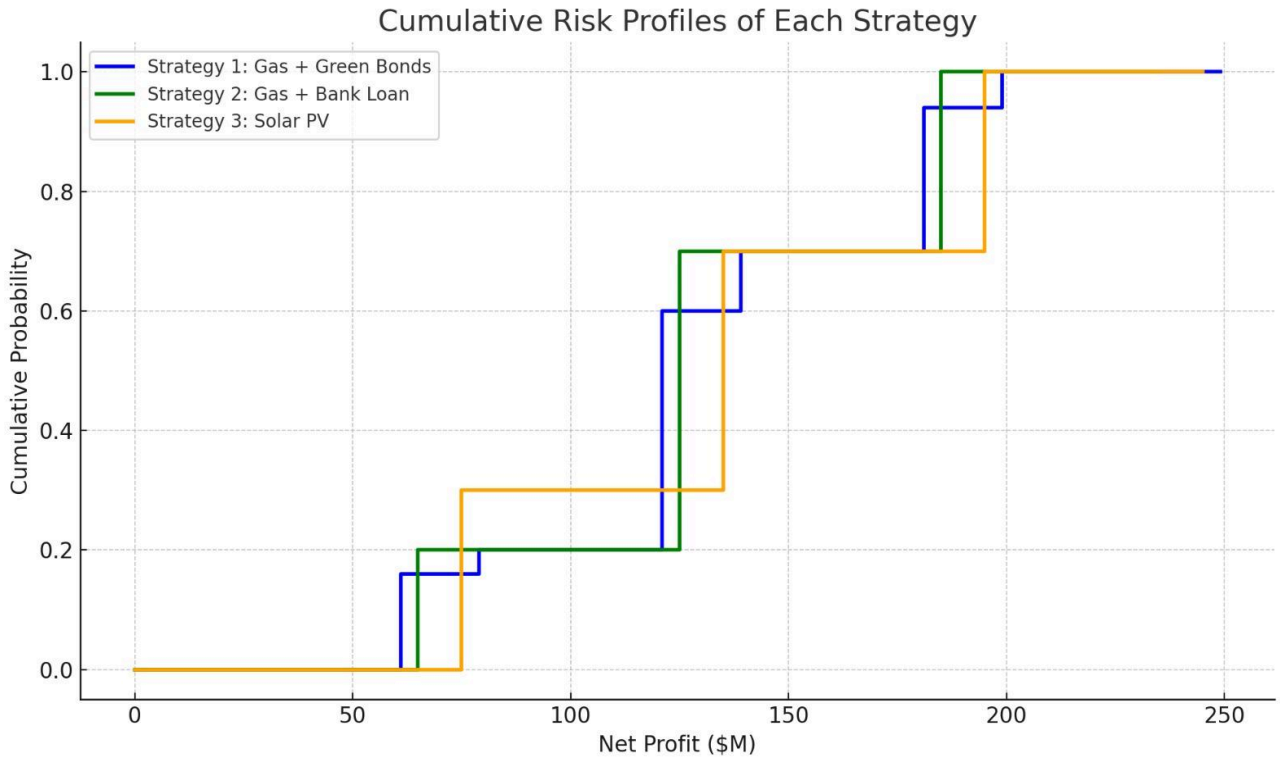


Figure 2.1. Cumulative risk profiles of each strategy

Interpretation of the CDFs:

Strategy 1 – Gas Plant + Green Bonds

Broadest fan of outcomes. Profits can be as low as ≈ \$60 M or as high as ≈ \$200 M. The CDF climbs slowly at first (16 % probability of ≤ \$61 M) and then levels off only near \$199 M, signalling both high downside risk and exceptional upside.

Strategy 2 – Gas Plant + Bank Loan

Moderate spread. Results lie between ≈ \$65 M and ≈ \$185 M. The CDF makes its biggest jump between \$65 M and \$125 M, meaning half of all outcomes sit tightly around the centre; extremes are narrower than in Strategy 1.

Strategy 3 – Solar PV Farm

Most compact downside, tempered upside. The worst case never falls below ≈ $75 M, and the best case stops at ≈ $195 M. The CDF rises steeply, so probability mass concentrates in a narrow band. Investors see a "safer but capped" profile.

Pairwise Dominance Analysis:

1. Strategy 1 vs Strategy 2:
    - Below roughly $70 M, Strategy 2 never dips lower than Strategy 1's minimum.
    - Between $120 M and $180 M, Strategy 2's CDF sits to the right (higher cumulative probability), again favouring the bank-loan option for middling outcomes.
    - Beyond $185 M, Strategy 1 suddenly pulls ahead, offering a chance of extreme upside that Strategy 2 cannot match.
      Result: the curves intersect, so no dominance.
2. Strategy 1 vs Strategy 3:
    - Strategy 3 dominates in the extreme low tail: its worst case is $75 M, whereas Strategy 1 can fall to $61 M.
    - From about $90 M up to $190 M, the two CDFs trade places; neither stays consistently to the right.
    - Above $195 M, only Strategy 1 retains probability mass, appealing to risk-seekers.
      Result: the curves intersect, so no dominance.

3. Strategy 2 vs Strategy 3:
    - At every percentile below $75 M, Strategy 3 is superior (Strategy 2 reaches $65 M).
    - Between $75 M and $135 M, the CDFs overlap almost perfectly.
    - From $135 M to $185 M, Strategy 2 edges ahead; above $195 M neither strategy offers additional upside.
      Result: curves intersect, so no dominance.

3. SENSITIVITY ANALYSIS

Even the best-crafted decision model rests on estimates: market prices, financing costs, capital budgets, demand forecasts. Sensitivity analysis asks, “How fragile is our recommendation if those numbers shift?”. By systematically perturbing the inputs and re-evaluating outcomes, we learn

- which assumptions drive the EMV gap between alternatives,
- where tipping points occur—values at which a different strategy becomes optimal, and
- how much managerial attention (hedging, contract design, data gathering) each uncertainty deserves.

For clarity we separate the investigation into one-way and two-way analyses.

3.1. ONE-WAY SENSITIVITY ANALYSIS

A one-way (or univariate) sweep alters one input at a time while holding all others at their base-case levels. The procedure for each critical variable is:

1. Select a plausible range – e.g., natural-gas prices ±30 %, bond coupon ±100 bp, PV module costs ±15 %.
2. Recompute the EMV (or EU for risk-adjusted views) of every strategy at each test point.
3. Plot the results as a tornado or line chart, revealing which variable causes the steepest slope and where crossover points emerge.

This single-factor lens highlights dominant drivers of project value and identifies straightforward break-even thresholds the board can monitor.

Parameter definition:

- p – probability that the green-bond market clears at a 3.8 % coupon.
- 1 − p – probability the issue must be placed at 4.2 %.
  All other inputs (gas prices, bank-loan rate, electricity prices) remain at their base-case values.

Strategy 1 as a function of p:

$$EMV_{S1}(p) = p * 145\ M + (1 - p) * 127\ M = 127 + 18p \text{ (in millions of dollars).}$$

Strategies 2 and 3 do not depend on p:

$$EMV_{S2} = 131\ M \text{ and } EMV_{S3} = 135\ M$$

When does Strategy 1 overtake Strategy 2?

$$127 + 18p > 131$$

$$p > 4/18 \approx 0.223$$

When does Strategy 1 overtake Strategy 3?

$$127 + 18p > 135$$

$$p > 8/18 \approx 0.445$$

If management believes the low-coupon outcome is very unlikely—below roughly one chance in five—green-bond financing is the weakest of the three options; the PV farm still dominates, with the bank-loan gas plant in second place.

As soon as the probability of securing 3.8 % rises above 22 % but stays below 45 %, the bond option leap-frogs the bank-loan variant but still trails Solar PV; it becomes a viable runner-up rather than the outright winner.

Only when decision makers are confident the favourable coupon is at least 45 % likely does Strategy 1 pull ahead of the PV project on expected monetary terms. In other words, the current recommendation (Solar PV) is quite robust: it would take more than doubling the base-case

probability (from 20 % to ≈ 45 %) for the gas-plant-plus-green-bonds strategy to become superior.

This threshold, p ≈ 0.445, is therefore a clear managerial trigger: unless intelligence from bond underwriters indicates odds close to—or better than—fifty–fifty for a 3.8 % coupon, switching away from Solar PV is not justified by EMV considerations.

3.2. TWO-WAY SENSITIVITY ANALYSIS

Parameters

- p represents the probability of low electricity price.
- q represents the probability of moderate electricity price.
- 1 − p − q represents the probability of high gas price.

Expected Monetary Value (EMV) Calculations:

E(S1) = (0.2×\$145.0 M) + (0.8×\$127.0 M) = \$29.0 M + \$101.6 M = \$130.6 M

E(S2) = (0.2×\$65 M) + (0.5×\$125 M) + (0.3×\$185 M) = \$131 M

E(S3)= (\$75 M×p) + (\$135 M×q) + (\$195M×(1−p−q)) = \$195 M - 120p - 60q

3.2.1. Comparison of Strategy 1 and Strategy 2

$$E(\textit{Strategy}\ 2) > E(\textit{Strategy}\ 1)$$

$$\$131\ M > \$130.6\ M$$

$$p + q \leq 1$$

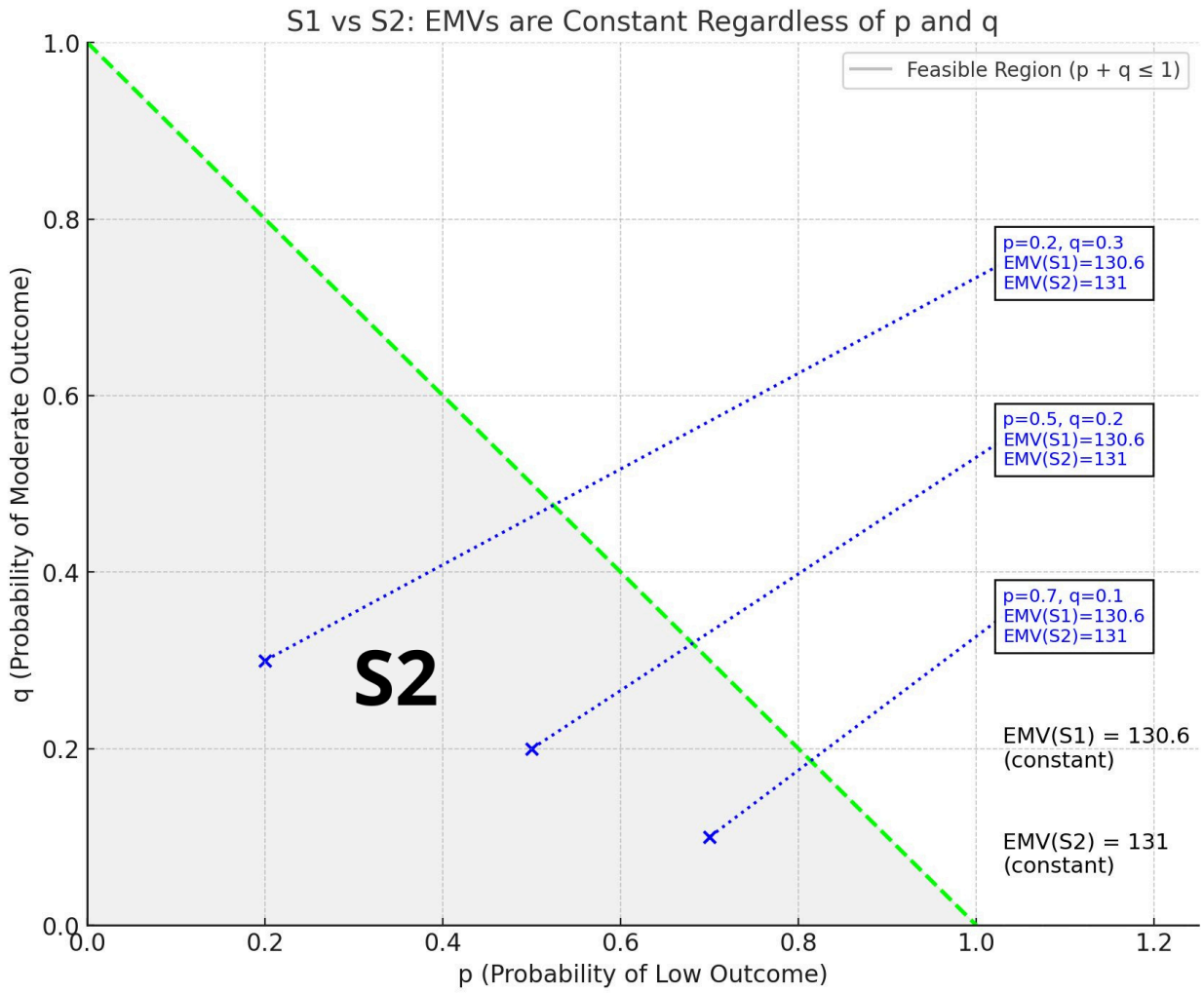


Figure 3.1. 2-way sensitivity analysis of strategy 1 and strategy 2

The two-way surface confirms that, for the gas-plant alternatives alone, electricity-price uncertainty simply does not matter: both EMV planes are flat, locked at $130.6 million for the green-bond option (S1) and $131 million for the bank-loan option (S2). No matter how the decision maker shifts belief between a low-price world (p) and a moderate-price world (q)—so long as the probabilities remain within the feasible triangle ($p + q \leq 1$)—the bank-loan strategy always edges out the green bond by the same $0.4 million. In practical terms, refining estimates of p and q will never reverse their ranking; if management wants to tilt the outcome toward green-bond financing, it must focus on variables that do affect S1's pay-off, such as the chance of securing the 3.8 % coupon or the underlying gas-price scenarios, rather than on the distribution of electricity prices.

3.2.2. Comparison of Strategy 1 and Strategy 3

$$E(\textit{Strategy } 1) > E(\textit{Strategy } 3)$$

$$\$130.6\ M > \$195M - 120p - 60q$$

$$q = 0 \rightarrow p \approx 0{,}5367$$

$$q = 1 \rightarrow p \approx 0{,}0367$$

$$p + q \leq 1$$

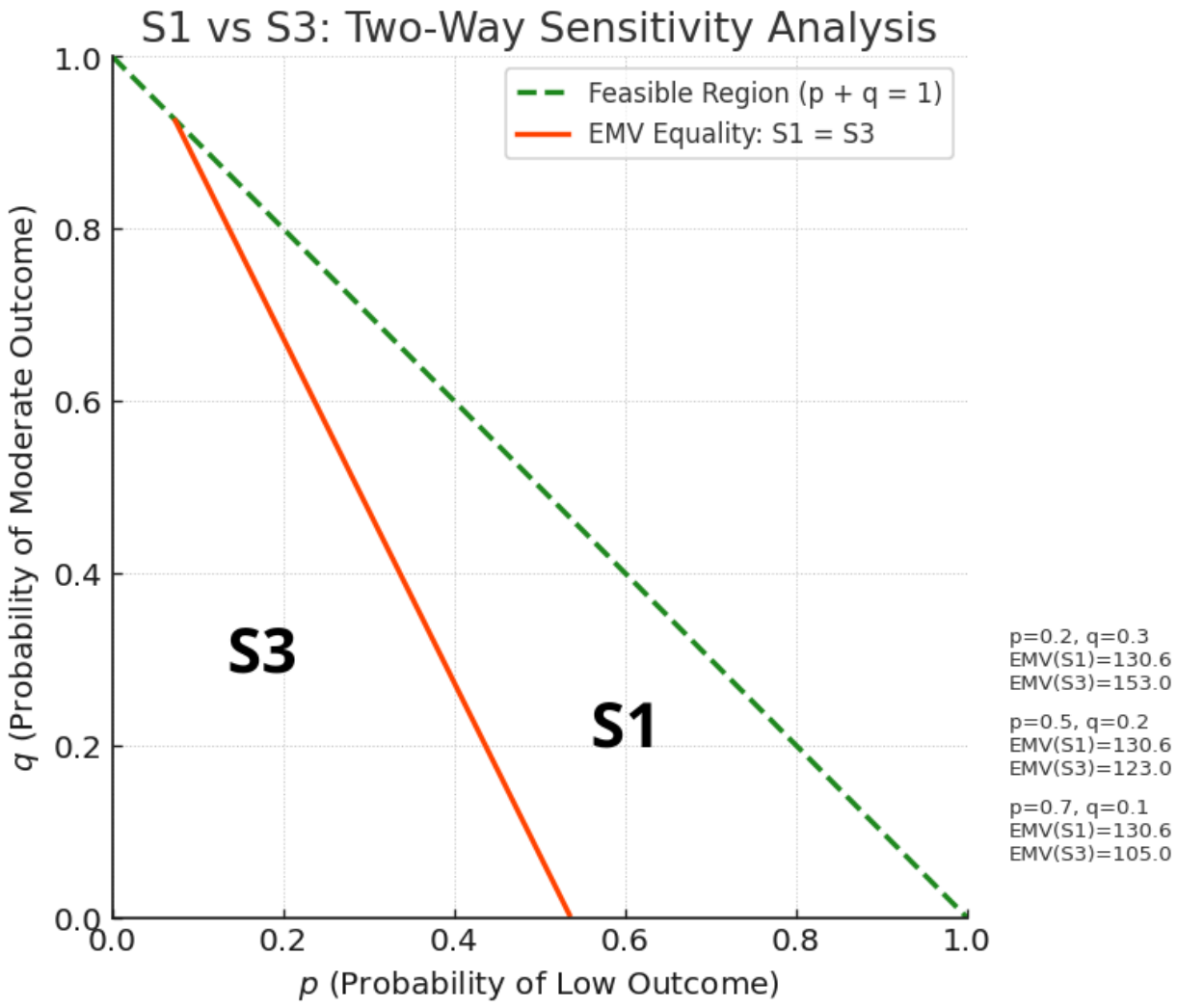


Figure 3.2. 2-way sensitivity analysis of strategy 1 and strategy 3

When more than roughly 54 % of the total probability mass is assigned to low and moderate electricity-price scenarios—leaving less than about 7 % for the high-price outcome—the green-bond gas plant edges ahead on expected profit, because the PV farm's upside is simply not triggered often enough to offset its higher capital outlay. As soon as even a modest slice of belief is shifted back to the high-price band, the solar project regains the lead; every extra 10-percentage-point chance of high prices lifts its EMV by \$60 million, whereas the gas plant's EMV is fixed. In practical terms, the PV farm remains the financially superior choice under any price forecast that is not overwhelmingly pessimistic.

3.2.3. Comparison of Strategy 2 and Strategy 3

$$\mathrm{E}(\textit{Strategy}\ 2) > E(\textit{Strategy}\ 3)$$

$$\$131\ \mathrm{M} > \$195\ \mathrm{M} - 120p - 60q$$

$$q = 0 \rightarrow p \approx\ 0{,}5333$$

$$q = 1 \rightarrow p \approx\ 0{,}0333$$

$$p + q \leq 1$$

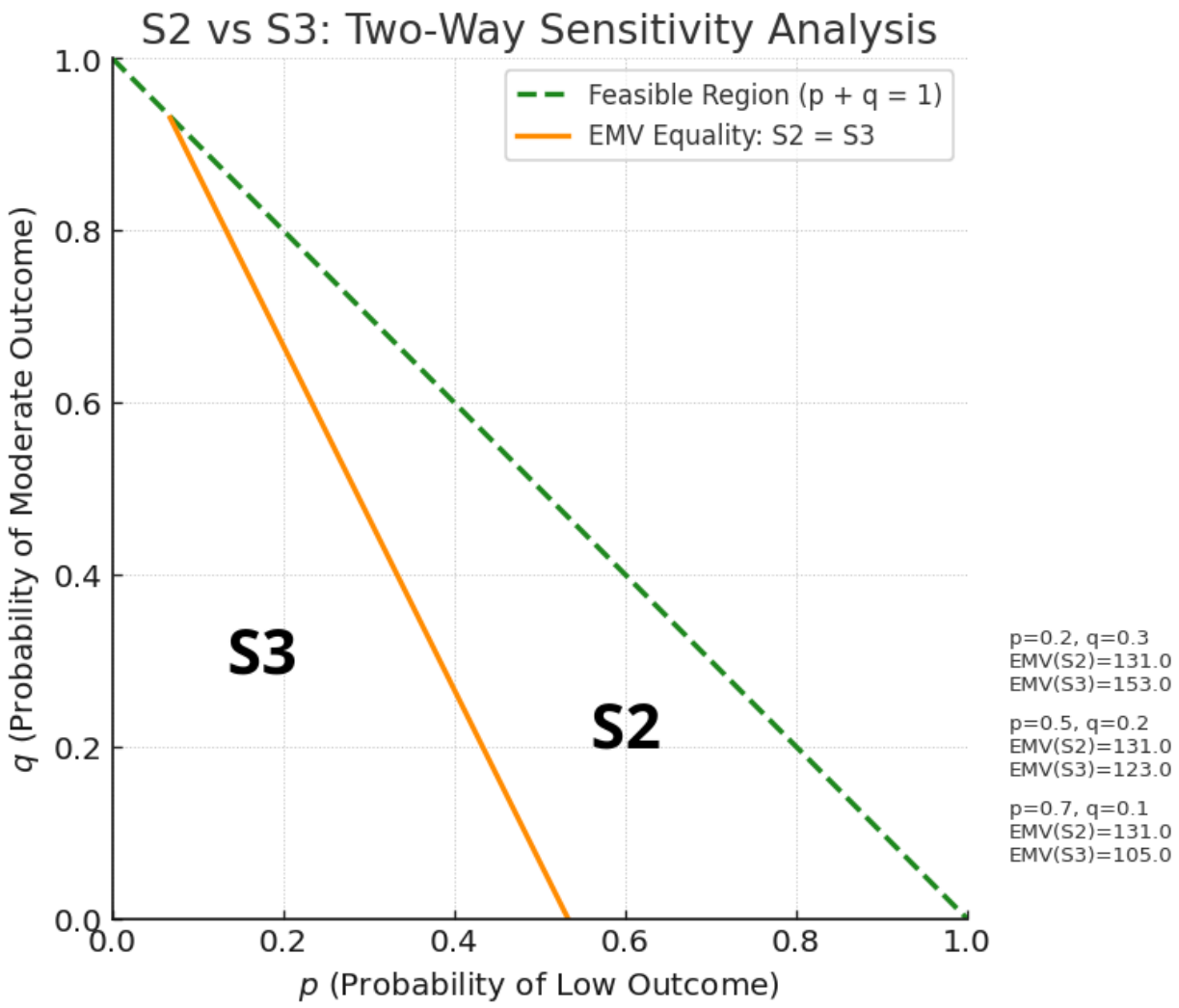


Figure 3.3. 2-way sensitivity analysis of strategy 2 and strategy 3

Replacing green bonds with a bank loan raises the gas plant's baseline EMV by $0.4 million, widening its favourable zone slightly, yet the story hardly changes: the gas plant becomes preferable only when the combined probability of low and moderate prices climbs above about 53 %. If management's outlook allows even a small (≈ 6–7 %) chance that wholesale prices will sit in the high band, the PV farm again delivers the larger expected return. Thus, unless long-range price forecasts turn decisively bearish, Solar PV continues to dominate the bank-loan gas alternative as well.

4. VALUE OF INFORMATION

When key uncertainties can be reduced or resolved before a commitment is made, a firm's decision tree changes: instead of locking into a single strategy today, management can condition its choice on the state of the world that eventually materialises.

Value of Information analysis quantifies the financial benefit of acquiring such knowledge, helping answer two practical questions:

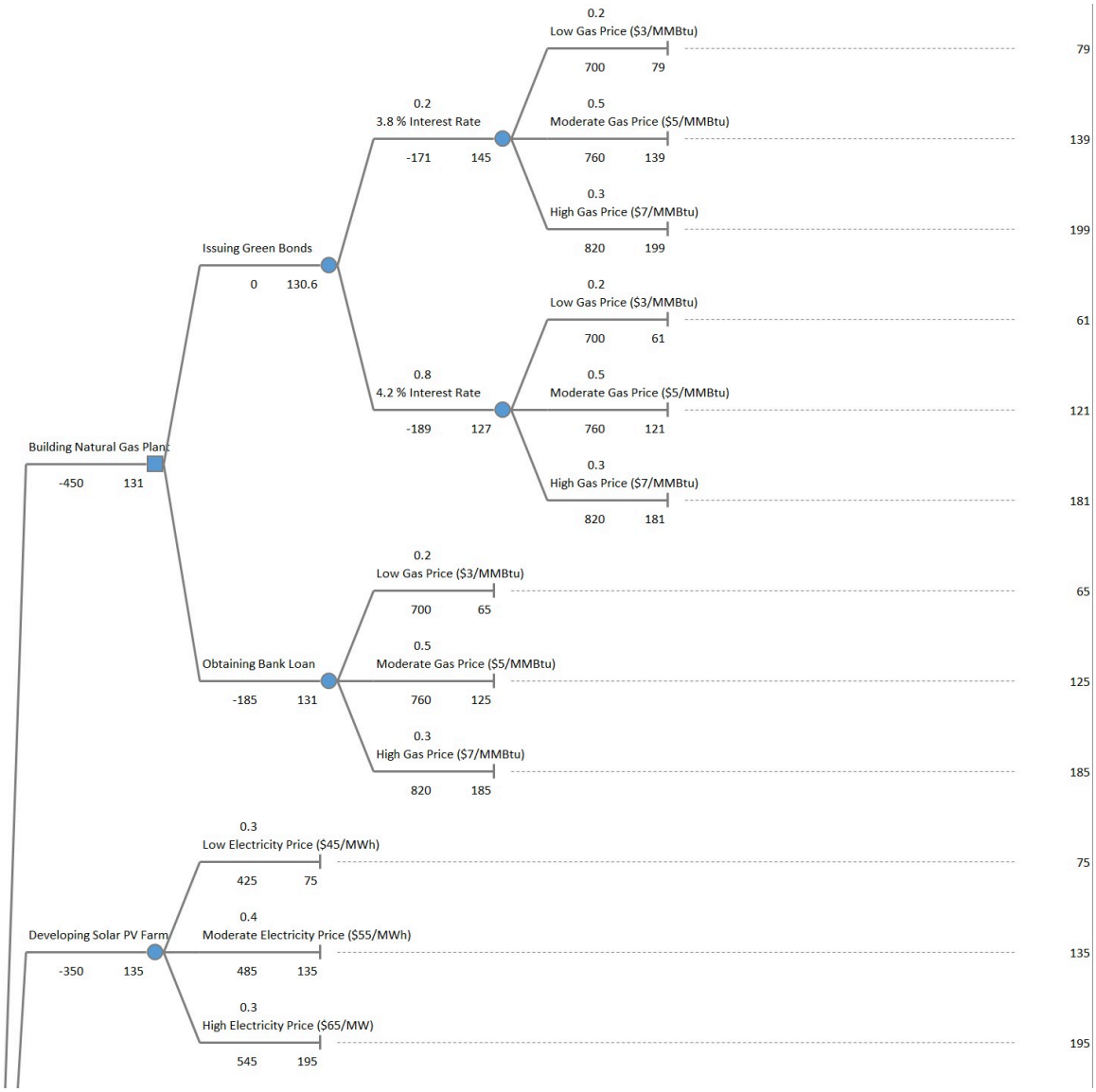
- Worth: What is the maximum GreenWave should be willing to pay for better forecasts, market studies, or sensor data?
- Focus: Which uncertainty is most lucrative to resolve?—gas prices, electricity prices, interest rates, or some other factor.

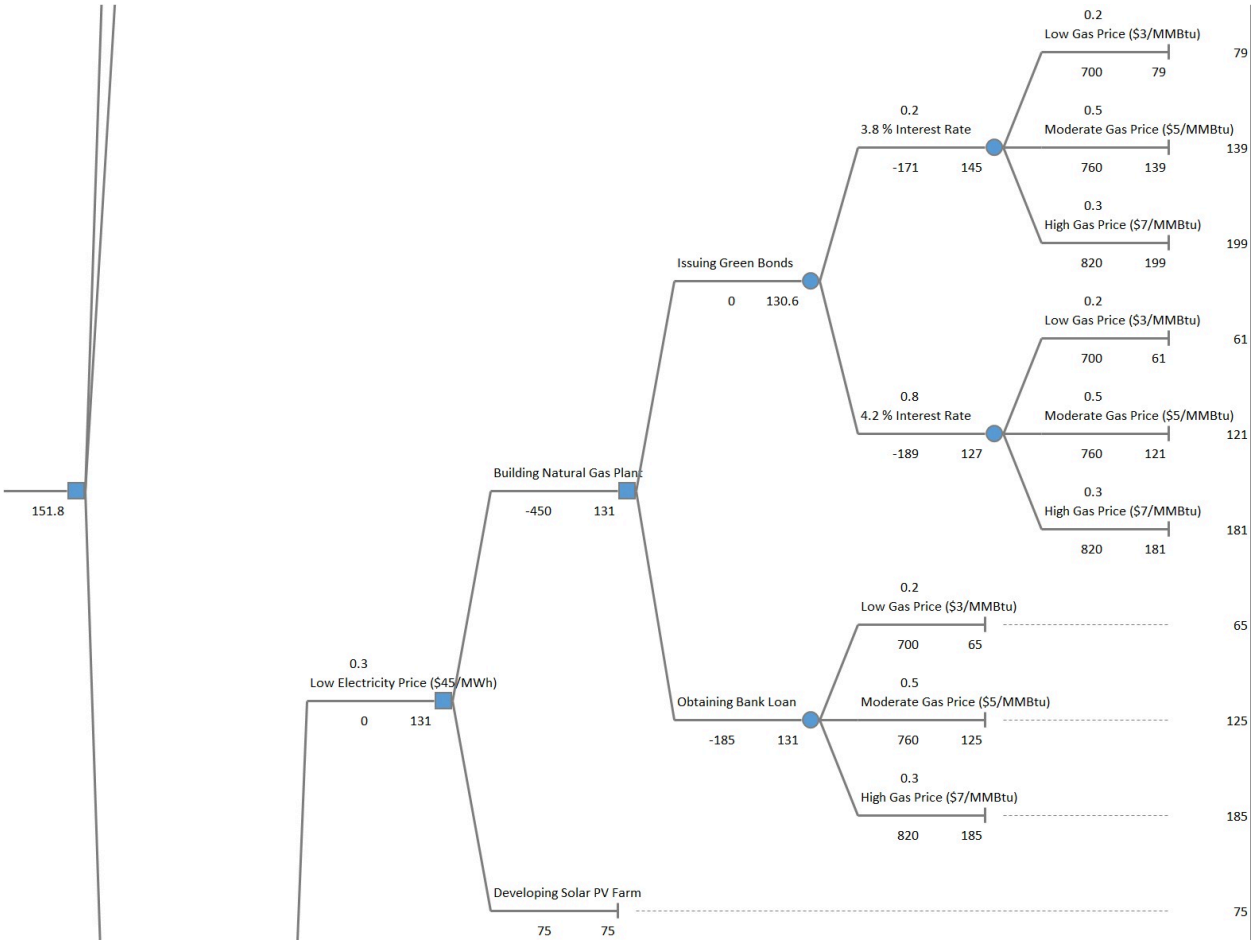
Two benchmark concepts anchor the analysis:

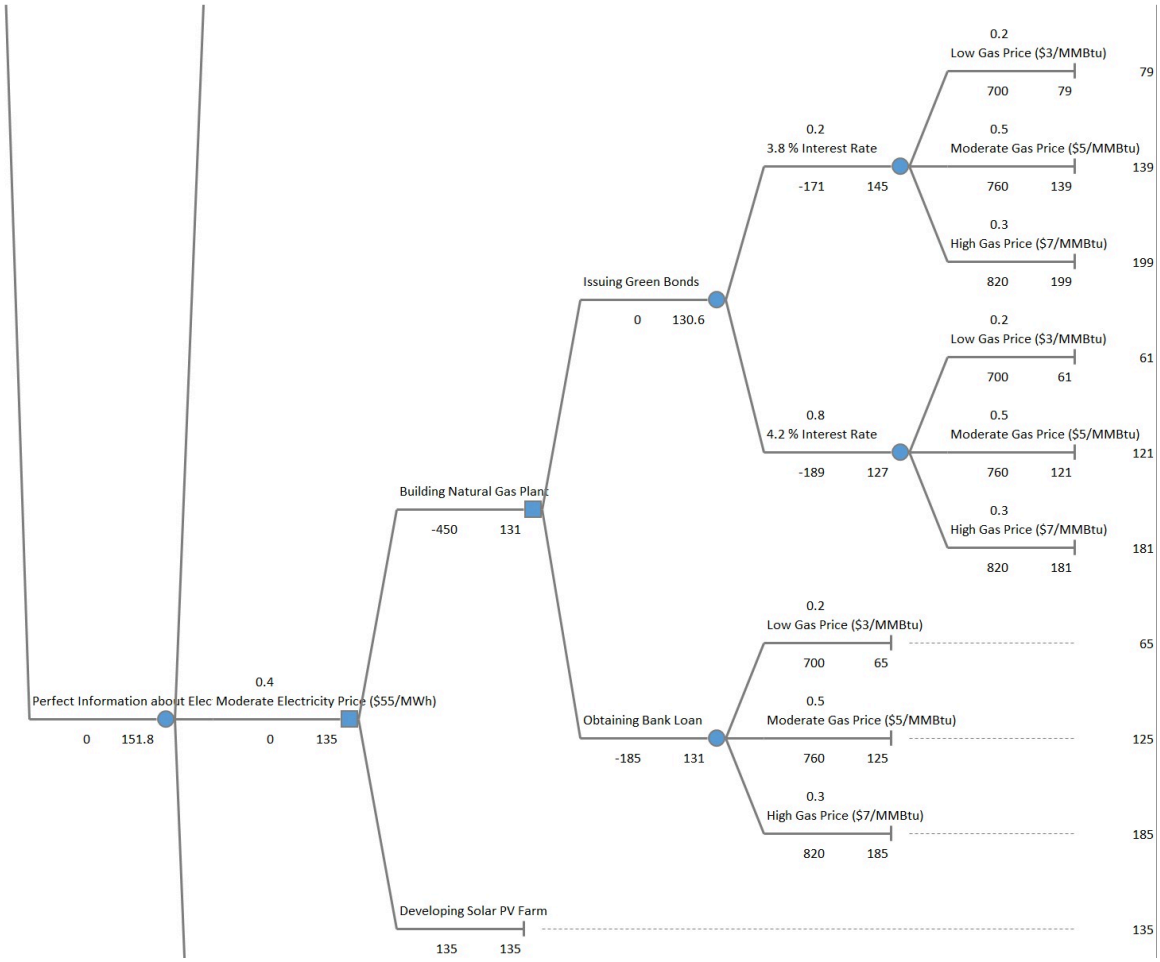
- Expected Value with Perfect Information (EVPI) – the average outcome if the true state becomes known for certain before the final decision.
- Expected Value of Perfect Information (EVII) – the extra profit attainable with perfect information.

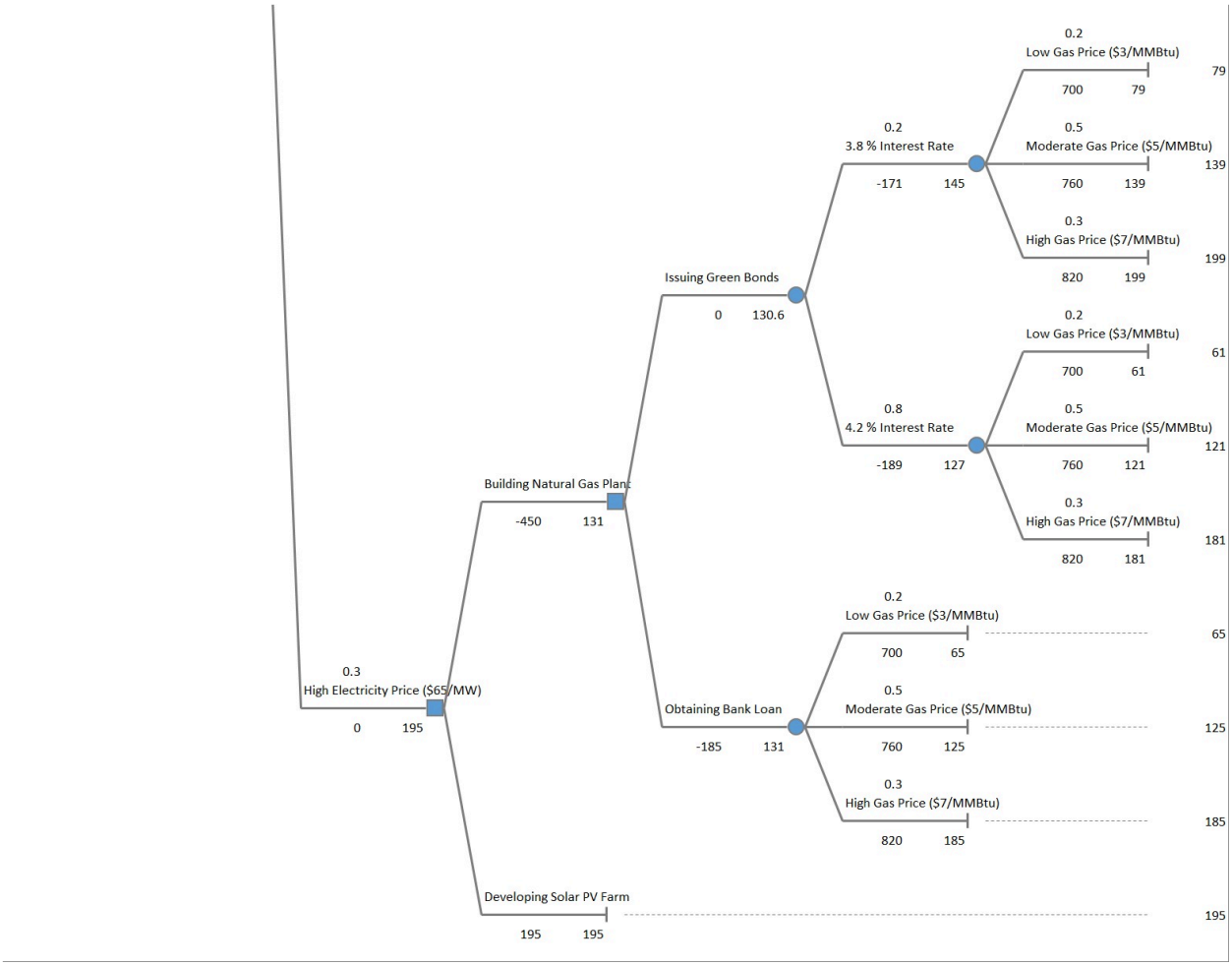
4.1. PERFECT INFORMATION ABOUT ELECTRICITY PRICE

We assume that, before committing capital, GreenWave receives a flawless forecast of the 2028–2038 wholesale electricity price band—Low ($45/MWh), Moderate ($55/MWh), or High ($65/MWh). Armed with that forecast, the company can pursue whichever generation option maximises net profit for that realised price scenario.


Building Natural Gas Plant
-450 131
Issuing Green Bonds
0 130.6
0.2
3.8 % Interest Rate
-171 145
0.2
Low Gas Price ($3/MMBtu)
700 79
79
0.5
Moderate Gas Price ($5/MMBtu)
760 139
139
0.3
High Gas Price ($7/MMBtu)
820 199
199
0.8
4.2 % Interest Rate
-189 127
0.2
Low Gas Price ($3/MMBtu)
700 61
61
0.5
Moderate Gas Price ($5/MMBtu)
760 121
121
0.3
High Gas Price ($7/MMBtu)
820 181
181
Obtaining Bank Loan
-185 131
0.2
Low Gas Price ($3/MMBtu)
700 65
65
0.5
Moderate Gas Price ($5/MMBtu)
760 125
125
0.3
High Gas Price ($7/MMBtu)
820 185
185
Developing Solar PV Farm
-350 135
0.3
Low Electricity Price ($45/MWh)
425 75
75
0.4
Moderate Electricity Price ($55/MWh)
485 135
135
0.3
High Electricity Price ($65/MW)
545 195
195


151.8
0.3
Low Electricity Price ($45/MWh)
0
131
Building Natural Gas Plant
-450
131
Developing Solar PV Farm
75
75
Issuing Green Bonds
0
130.6
Obtaining Bank Loan
-185
131
0.2
3.8 % Interest Rate
-171
145
0.8
4.2 % Interest Rate
-189
127
0.2
Low Gas Price ($3/MMBtu)
700
79
79
0.5
Moderate Gas Price ($5/MMBtu)
760
139
139
0.3
High Gas Price ($7/MMBtu)
820
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199
0.2
Low Gas Price ($3/MMBtu)
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61
0.5
Moderate Gas Price ($5/MMBtu)
760
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0.3
High Gas Price ($7/MMBtu)
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Low Gas Price ($3/MMBtu)
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Moderate Gas Price ($5/MMBtu)
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High Gas Price ($7/MMBtu)
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Perfect Information about Elec
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Moderate Electricity Price ($55/MWh)
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Building Natural Gas Plant
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131
Developing Solar PV Farm
135
135
Issuing Green Bonds
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130.6
Obtaining Bank Loan
-185
131
0.2
3.8 % Interest Rate
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145
0.8
4.2 % Interest Rate
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Low Gas Price ($3/MMBtu)
700
79
79
0.5
Moderate Gas Price ($5/MMBtu)
760
139
139
0.3
High Gas Price ($7/MMBtu)
820
199
199
0.2
Low Gas Price ($3/MMBtu)
700
61
61
0.5
Moderate Gas Price ($5/MMBtu)
760
121
121
0.3
High Gas Price ($7/MMBtu)
820
181
181
0.2
Low Gas Price ($3/MMBtu)
700
65
65
0.5
Moderate Gas Price ($5/MMBtu)
760
125
125
0.3
High Gas Price ($7/MMBtu)
820
185
185



Knowing the exact electricity-price regime elevates GreenWave's expected net profit from $135 M to $151.8 M—an increase of $16.8 M. This improvement arises because:

1. Adaptive strategy selection:

    - If prices are Low or Moderate, the natural-gas plant financed by a bank loan yields the higher payoff.
    - If prices are High, the Solar PV farm becomes superior.

      By switching strategies in this contingent manner, GreenWave captures upside in each scenario while avoiding mismatches between technology and market conditions.

2. Risk-profile enhancement:

   Perfect information truncates downside outcomes: the company no longer faces the risk of committing to the Solar PV project only to discover later that electricity prices stayed near $45/MWh. Expected profits therefore rise not through "free money," but through better alignment between investment and prevailing prices.

3. Monetary yardstick for data spending:

   The EVPI of $16.8 M represents an upper limit on what GreenWave should ever pay—collectively—for market intelligence that pins down long-term electricity prices. Any price-forecasting study, option contract, or strategic partnership costing less than $16.8 M and delivering equivalent certainty would add shareholder value; anything more would erode it.

4. Strategic insight:

   - The relatively modest EVPI (about 12 % of baseline EMV) indicates that the current recommendation (Solar PV without extra information) is already robust.
   - However, electricity-price uncertainty still outweighs interest-rate uncertainty in its impact, because removing the latter would not change the ability to pivot between technologies.

Perfect foresight on electricity prices would make GreenWave $16.8 M richer in expected terms, but the company must weigh that theoretical benefit against the real cost and feasibility of obtaining such flawless information.

4.2. IMPERFECT INFORMATION ABOUT ELECTRICITY PRICE

P("Low" | Low) = 0.8

P("Moderate" | Low) = 0.1

P("High" | Low) = 0.1

P("Low" | Moderate) = 0.2

P("Moderate" | Moderate) = 0.6

P("High" | Moderate) = 0.2

P("Low" | High) = 0.1

P("Moderate" | High) = 0.3

P("High" | High) = 0.6

P(Low)=0.3

P(Moderate)=0.4

P(High)=0.3

P("Low") = P("Low" | Low) * P(Low) + P("Low" | Moderate) * P(Moderate) + P("Low" | High) * P(High) = 0.8 * 0.3 + 0.1 * 0.3 + 0.2 * 0.4 = 0.24 + 0.03 + 0.08 = 0.35

P("Moderate") = P("Moderate" | Low) * P(Low) + P("Moderate" | Moderate) * P(Moderate) + P("Moderate" | High) * P(High) = 0.1 * 0.3 + 0.3 * 0.3 + 0.6 * 0.4 = 0.03 + 0.09 + 0.24 = 0.36

P("High") = P("High" | Low) * P(Low) + P("High" | Moderate) * P(Moderate) + P("High" | High) * P(High) = 0.1 * 0.3 + 0.6 * 0.3 + 0.2 * 0.4 = 0.03 + 0.18 + 0.08 = 0.29

P("Low") + P("Moderate") +P("High") = 0.35 + 0.36 + 0.29 = 1

P(Low | "Low") = [P("Low" | Low) * P(Low)] / P("Low") = (0.8 * 0.3) / 0.35 = 0.686

P(Moderate | "Low") = [P("Low" | Moderate) * P(Moderate)] / P("Low") = (0.2 * 0.4) / 0.35 = 0.228

P(High | "Low") = [P("Low" | High) * P(High)] / P("Low") = (0.1 * 0.3) / 0.35 = 0.086

P(Low | "Low") + P(Moderate | "Low") + P(High | "Low") = 0.686 + 0.228 + 0.086 = 1

P(Low | "Moderate") = P("Moderate" | Low) * P(Low) / P("Moderate") = (0.1 * 0.3) / 0.36 = 0.083

P(Moderate | "Moderate") = P("Moderate" | Moderate) * P(Moderate) / P("Moderate") = (0.6 * 0.4) / 0.36 = 0.667

P(High | "Moderate") = P("Moderate" | High) * P(High) / P("Moderate") = (0.3 * 0.3) / 0.36 = 0.25

P(Low | "Moderate") + P(Moderate | "Moderate") + P(High | "Moderate") = 0.083 + 0.667 + 0.25 = 1

P(Low | "High") = [P("High" | Low) * P(Low)] / P("High") = (0.1 * 0.3) / 0.29 = 0.103

P(Moderate | "High") = [P("High" | Moderate) * P(Moderate)] / P("High") = (0.2 * 0.4) / 0.29 = 0.276

P(High | "High") = [P("High" | High) * P(High)] / P("High") = (0.6 * 0.3) / 0.29 = 0.621

P(Low | "High") + P(Moderate | "High") + P(High | "High") = 0.103 + 0.276 + 0.621 = 1

P(Low ∩ "Low") = P(Low | "Low") * P("Low") = 0.686 * 0.35 = 0.24

P(Moderate ∩ "Low") = P(Moderate | "Low") * P("Low") = 0.229 * 0.35 = 0.08

P(High ∩ "Low") = P(High | "Low") * P("Low") = 0.086 * 0.35 = 0.0301

P(Low ∩ "Moderate") = P(Low | "Moderate") * P("Moderate") = 0.083 * 0.36 = 0.0299

P(Moderate ∩ "Moderate") = P(Moderate | "Moderate") * P("Moderate") = 0.667 * 0.36 = 0.24

P(High ∩ "Moderate") = P(High | "Moderate") * P("Moderate") = 0.25 * 0.36 = 0.09

P(Low ∩ "High") = P(Low | "High") * P("High") = 0.103 * 0.29 = 0.03

P(Moderate ∩ "High") = P(Moderate | "High") * P("High") = 0.276 * 0.29 = 0.08

P(High ∩ "High") = P(High | "High") * P("High") = 0.621 * 0.29 = 0.18

Setup and signal quality:

GreenWave now receives an imperfect forecast—a three-level signal ("Low", "Moderate", "High") whose accuracy is given in the conditional matrix you supplied. The signal is informative but fallible: for example, when prices will actually be "Low", the model still reports "Moderate" or "High" 20 % of the time. Prior probabilities remain 0.30 / 0.40 / 0.30.

Posterior beliefs:

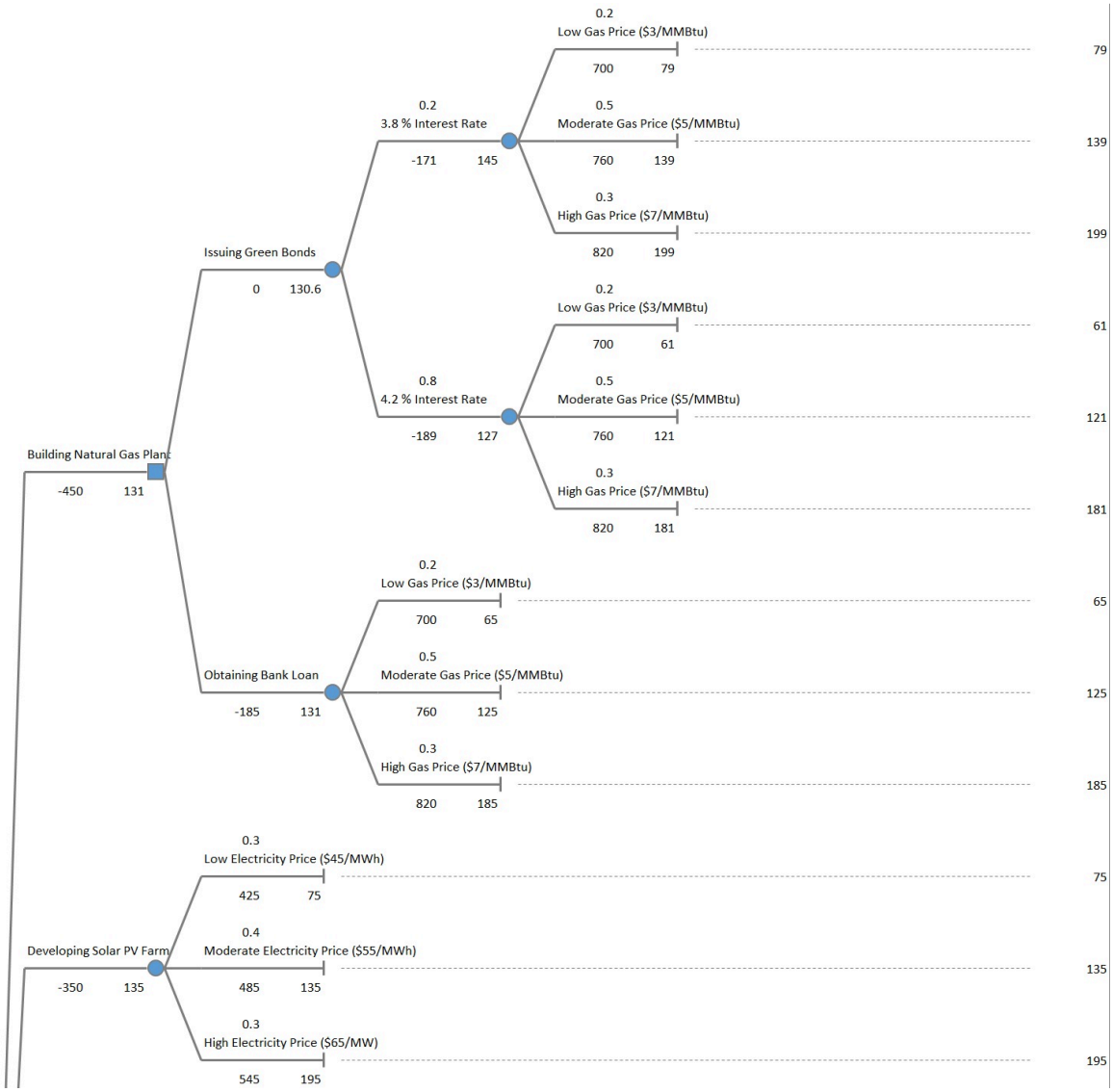
Using Bayes' rule (values already derived in the report), management can translate any observed signal into posterior odds—for instance, after a "Low" signal there is a 68.6 % chance that prices really will be in the low band, 22.8 % chance of moderate, and only 8.6 % chance of high. Similar posteriors follow for "Moderate" and "High".

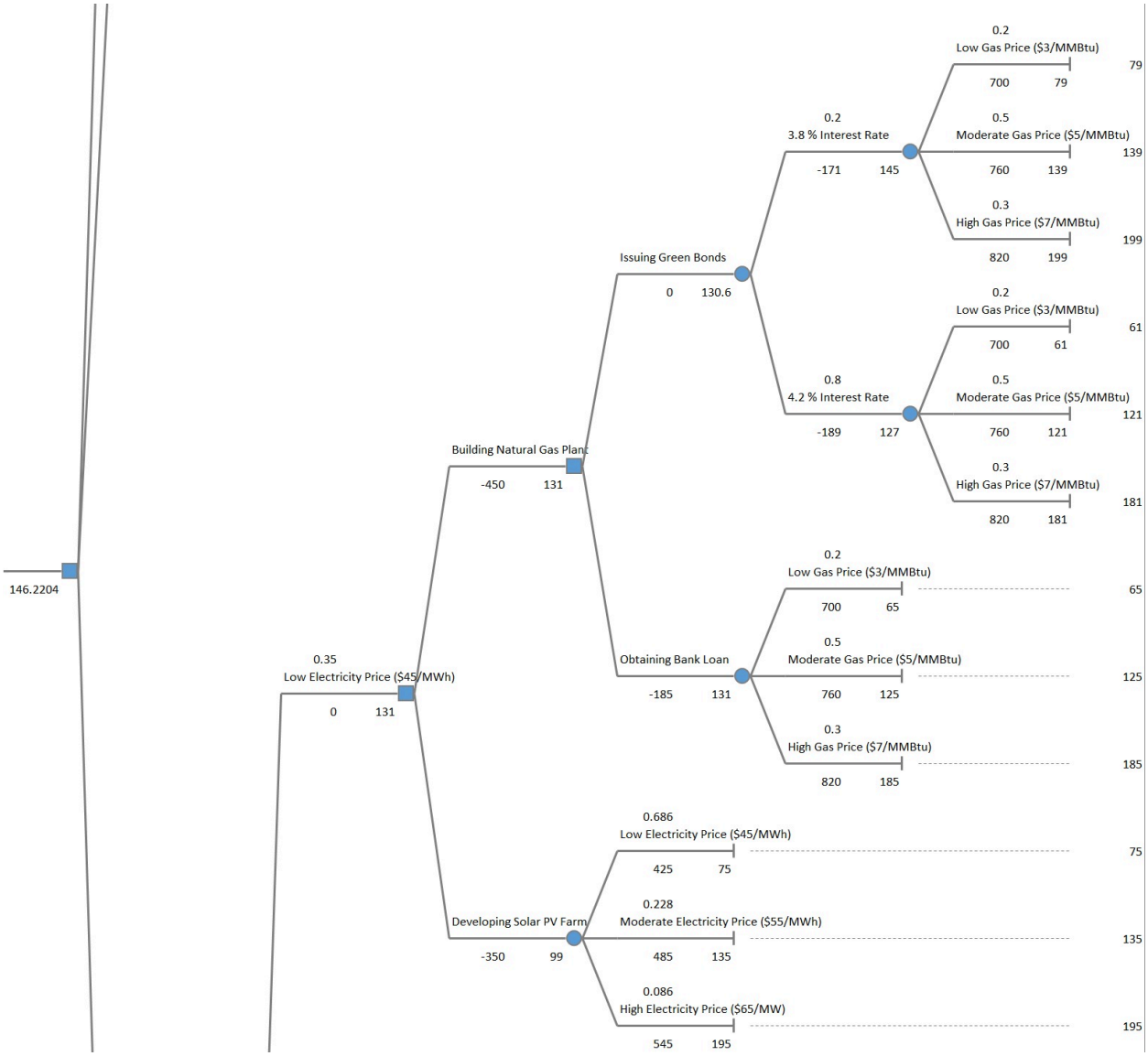
Contingent decision rule:

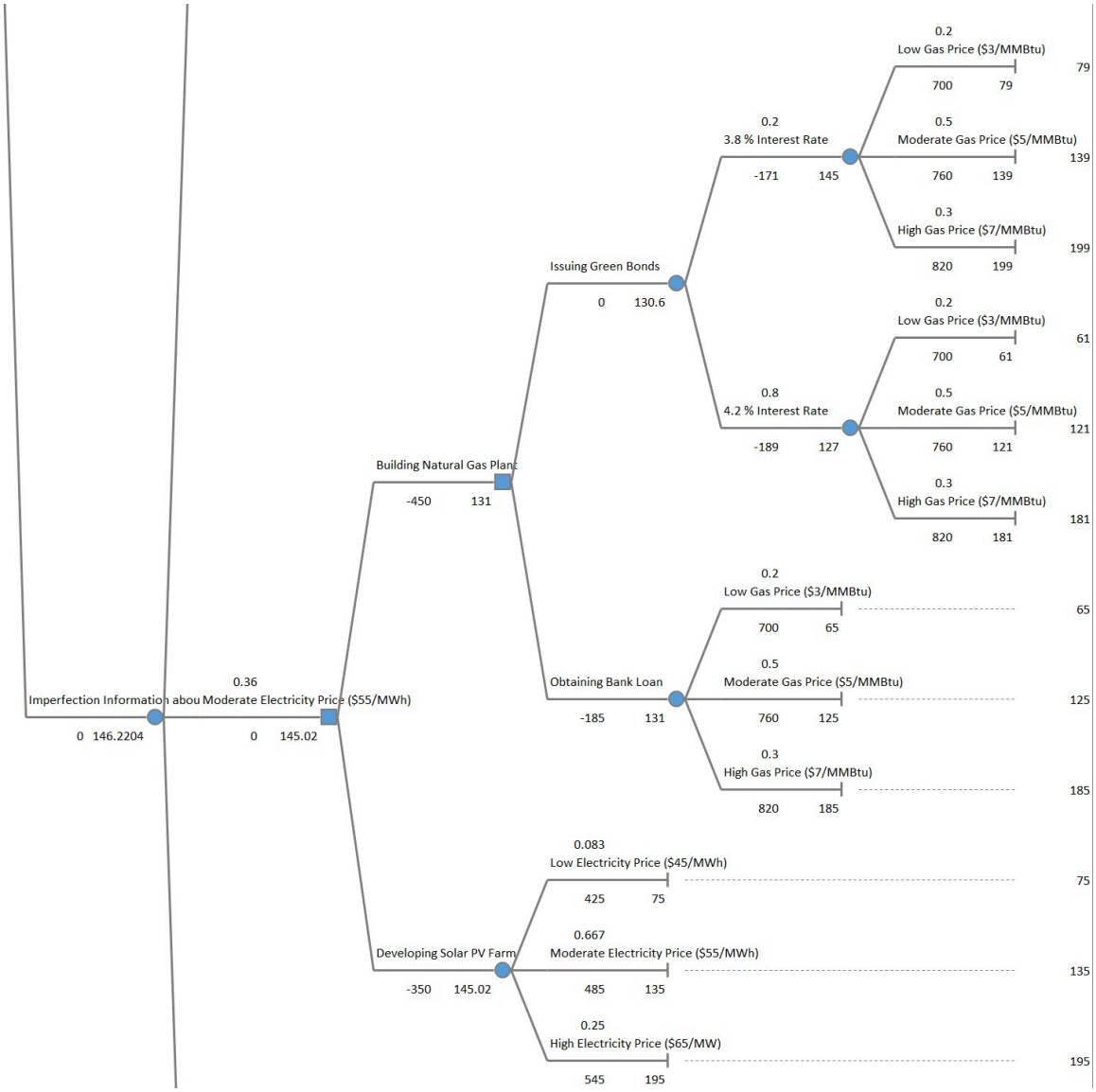
For each possible signal we recompute which project maximises conditional EMV:

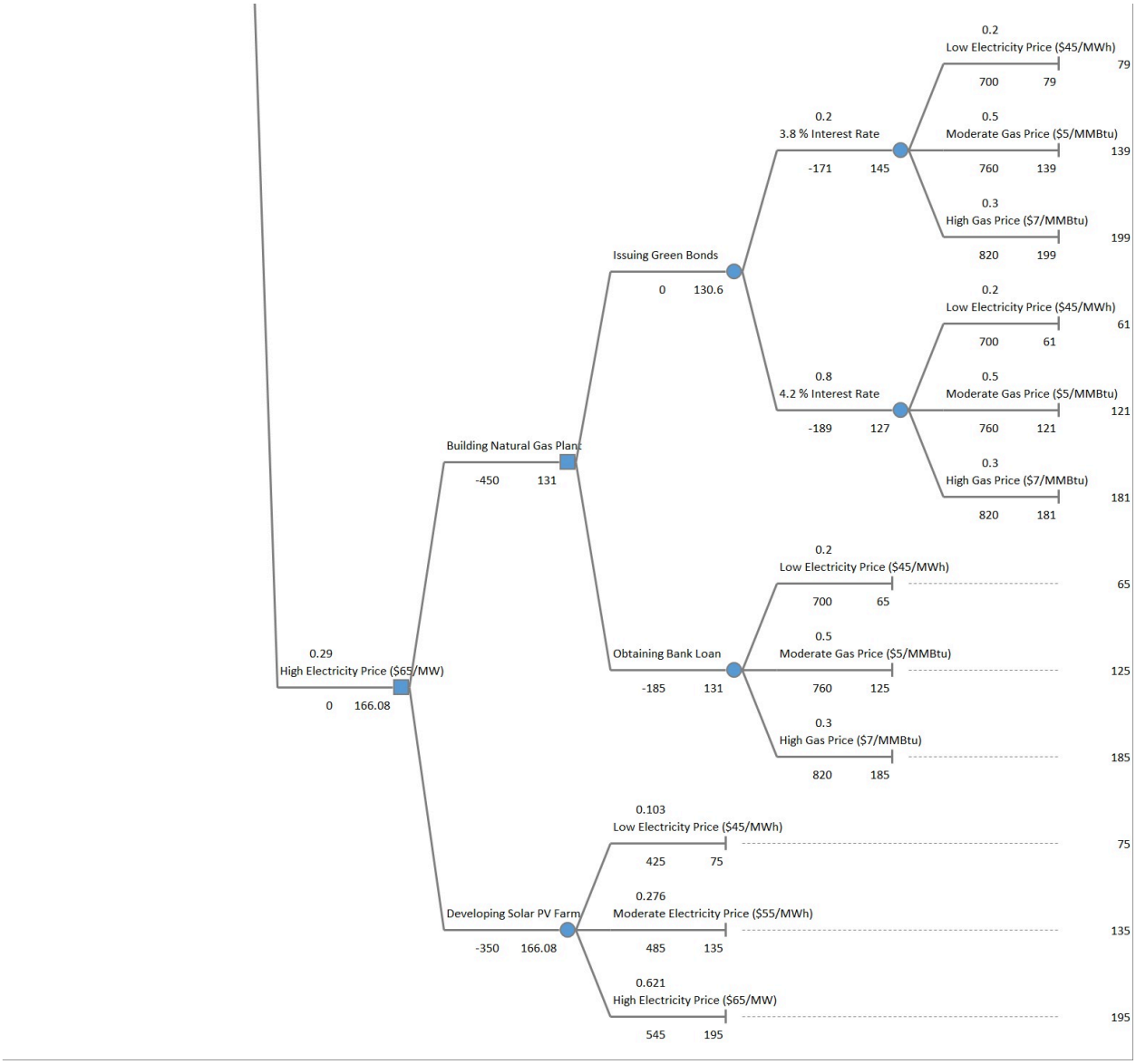
- Signal “Low” → bank-financed gas plant remains preferred (higher posterior EMV than Solar PV).
- Signal “Moderate” → bank-financed gas plant is again superior.
- Signal “High” → Solar PV farm overtakes the gas option thanks to the meaningful probability (62.1 %) of the truly high-price regime.

Thus the firm adopts a mixed strategy: pursue natural gas in two signals and solar in the third.




146.2204
0.35
Low Electricity Price ($45/MWh)
0
131
Building Natural Gas Plant
-450
131
Issuing Green Bonds
0
130.6
0.2
3.8 % Interest Rate
-171
145
0.2
Low Gas Price ($3/MMBtu)
700
79
79
0.5
Moderate Gas Price ($5/MMBtu)
760
139
139
0.3
High Gas Price ($7/MMBtu)
820
199
199
0.8
4.2 % Interest Rate
-189
127
0.2
Low Gas Price ($3/MMBtu)
700
61
61
0.5
Moderate Gas Price ($5/MMBtu)
760
121
121
0.3
High Gas Price ($7/MMBtu)
820
181
181
Obtaining Bank Loan
-185
131
0.2
Low Gas Price ($3/MMBtu)
700
65
65
0.5
Moderate Gas Price ($5/MMBtu)
760
125
125
0.3
High Gas Price ($7/MMBtu)
820
185
185
Developing Solar PV Farm
-350
99
0.686
Low Electricity Price ($45/MWh)
425
75
75
0.228
Moderate Electricity Price ($55/MWh)
485
135
135
0.086
High Electricity Price ($65/MW)
545
195
195


Imperfection Information abou
0 146.2204
0.36
Moderate Electricity Price ($55/MWh)
0 145.02
Building Natural Gas Plant
-450 131
Issuing Green Bonds
0 130.6
0.2
3.8 % Interest Rate
-171 145
0.2
Low Gas Price ($3/MMBtu)
700 79
79
0.5
Moderate Gas Price ($5/MMBtu)
760 139
139
0.3
High Gas Price ($7/MMBtu)
820 199
199
0.8
4.2 % Interest Rate
-189 127
0.2
Low Gas Price ($3/MMBtu)
700 61
61
0.5
Moderate Gas Price ($5/MMBtu)
760 121
121
0.3
High Gas Price ($7/MMBtu)
820 181
181
Obtaining Bank Loan
-185 131
0.2
Low Gas Price ($3/MMBtu)
700 65
65
0.5
Moderate Gas Price ($5/MMBtu)
760 125
125
0.3
High Gas Price ($7/MMBtu)
820 185
185
Developing Solar PV Farm
-350 145.02
0.083
Low Electricity Price ($45/MWh)
425 75
75
0.667
Moderate Electricity Price ($55/MWh)
485 135
135
0.25
High Electricity Price ($65/MW)
545 195
195



Knowing the exact electricity-price regime elevates GreenWave's expected net profit from $135 M to $151.8 M—an increase of $16.8 M. This improvement arises because:

1. Incremental gain over the status quo:

   Baseline EMV without extra information was $135 M. Imperfect information raises this to $146.22 M, an uplift of $11.22 M—the Expected Value of Imperfect Information (EVII).

2. Comparison with perfect information:

   Perfect foresight was worth \$16.8 M. The current forecast therefore captures roughly two-thirds of the theoretical maximum (≈ 67 %), leaving \$5.6 M still on the table due to residual misclassification risk.

3. Economic guideline for data spending:

   \$11.22 M is an upper ceiling on what GreenWave should collectively pay for this particular forecasting service, data subscription, or internal analytics effort. Any contract cost below that figure is value-adding; any cost above erodes shareholder value.

4. Strategic robustness:

   Even with noisy signals, the decision rule continues to select the Solar PV farm whenever upside is greatest, while defaulting to the gas-plant-and-loan combination in less favourable price scenarios. This adaptive posture narrows downside dispersion compared with a single static commitment.

5. Next steps:

   - Assess whether additional, complementary data (e.g., forward contracts, regional demand indicators) could tighten the signal matrix enough to approach the perfect-information benchmark.
   - Examine the cost of hedging instruments: if a modest hedge can replicate the \$11.22 M benefit more cheaply, it may replace or supplement further data collection.

The proposed imperfect-information forecast delivers a meaningful but not overwhelming improvement; management should treat \$11.22 M as the break-even reference point when negotiating data-acquisition or analytic-tool budgets.

5. UTILITY FUNCTIONS

Traditional expected monetary value (EMV) analysis treats every dollar as equally valuable no matter who receives it or how large the stake is. In practice, however, investors care not only about how much they might earn but also how uncertain those earnings are. A $10 million swing means something very different to a start-up than to a national utility.

Utility theory provides the bridge between raw cash flows and the decision maker's subjective preferences. Under the *Von Neumann–Morgenstern* axioms, each monetary consequence x is transformed by a utility function U(x) into a scale on which:

1. Risk attitudes are encoded:
    - Concave U(x) → diminishing marginal satisfaction → risk-averse behaviour.
    - Linear U(x) → constant marginal satisfaction → risk-neutral behaviour.
    - Convex U(x) → increasing marginal satisfaction → risk-seeking behaviour.
2. Choices are ranked by expected utility (EU), not by EMV.

$$EU = \sum_{s} P(s)\, U(x_s)$$

   The strategy with the highest EU is preferred, even if its EMV is smaller.

3. Certainty equivalents (CE) and risk premia (RP) become computable.
    - CE is the guaranteed cash amount that yields the same utility as the risky prospect.
    - RP = EMV − CE quantifies how much value the decision maker forgoes (if positive) or is willing to pay (if negative) to escape or embrace risk.

Risk-Averse Decision Maker:

- Utility: $U(x) = \ln(x)$ (natural logarithm, defined for $x>0$)
- Economic meaning: Diminishing marginal utility of wealth; every extra dollar adds less satisfaction than the previous one.
- Risk attitude: Prefers safer, lower-variance outcomes—even at the cost of some expected value—because the disutility of a large loss outweighs the utility of an equally large gain.

Risk-Neutral Decision Maker:

- Utility: $U(x) = 5x+7$ (any positive linear transformation is equivalent)
- Economic meaning: Constant marginal utility; each additional dollar is valued exactly the same, regardless of current wealth.
- Risk attitude: Indifferent to uncertainty. Choices are made solely to maximise expected monetary value; EU and EMV coincide.

Risk-Seeking Decision Maker:

- Utility: $U(x) = ½*x^2$ (quadratic, convex)
- Economic meaning: Increasing marginal utility; larger pay-offs create disproportionately higher satisfaction.
- Risk attitude: Actively favours high-variance prospects, willing to accept lower expected value for a chance at much larger gains.

These three canonical forms span the spectrum of risk preferences commonly analysed in energy-investment problems. Using all three lets us check whether the recommended strategy (currently the Solar PV farm) remains optimal under different attitudes toward risk—and, if not, how the preferred choice shifts as decision-maker psychology changes.

5.1. RISK-AVERSE DECISION MAKER

To represent a risk-averse decision maker, we adopt the concave utility function

$$U(x) = \ln(x)$$

a widely used specification that embeds diminishing marginal utility: each additional dollar yields progressively less satisfaction than the previous one. The natural-log form is analytically convenient, exhibits decreasing absolute risk aversion (DARA), and faithfully captures a preference for more certain—even if slightly smaller—returns over riskier alternatives with the same expected value.

Table 5.1. Utilities of the consequences based on the utility function of U(x) = ln(x)

x	U(x) = ln(x)
79	4,369447852
139	4,934473933
199	5,293304825
61	4,110873864
121	4,795790546
181	5,198497031
65	4,17438727
125	4,828313737
185	5,220355825
75	4,317488114
135	4,905274778
195	5,272999559

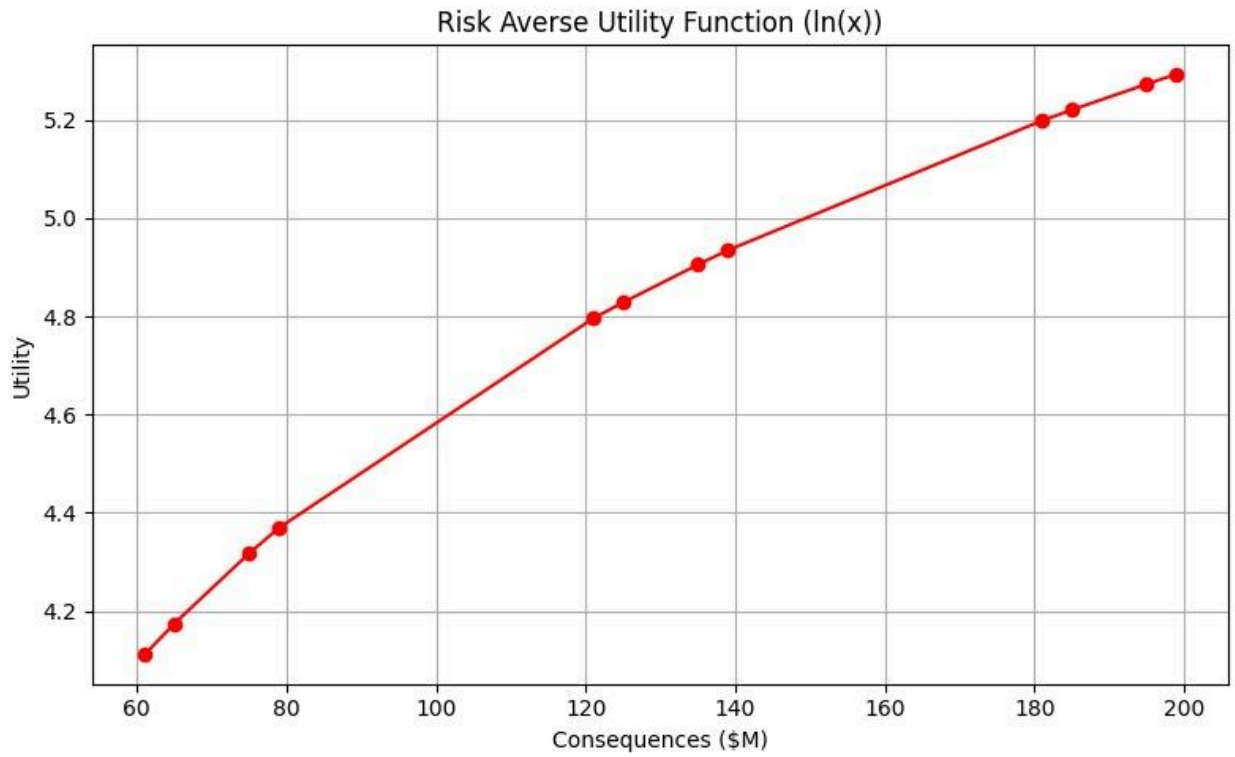


Figure 5.1. Plot of the utilities of the consequences based on the utility function U(x) = ln(x)

Figure 5.1 shows that U(x) = ln(x) is concave, signalling diminishing marginal utility—each additional dollar provides less satisfaction than the previous one. A decision-maker with this utility willingly forgoes part of the expected monetary value to reduce outcome variability, favouring certain, moderate pay-offs over riskier alternatives with the same expected return.

$$\rho = \frac{-U''(x)}{U'(x)}$$

U’(x) = 1/x

U’’(x) = $-1/x^2$

Coefficient of Risk Aversion = $- (-1/x^2) / (1/x) = 1/x$ (DARA)

U(3.8 Interest Rate) = 4.369 * 0.2 + 4.934 * 0.5 + 5.293 * 0.3 = 4.9287

U(4.2 Interest Rate) = 4.110 * 0.2 + 4.795* 0.5 + 5.198 * 0.3 = 4.7789

U(Issuing Green Bonds) = 4.9287 * 0.2 + 4.7789 * 0.8 = 4.80886

U(Obtaining Bank Loan) = 4.174 * 0.2 + 4.828 * 0.5 + 5.220 * 0.3 = 4.8148

U(Building Natural Gas Plant) = max(4.80886, 4.8148) = 4.8148

U(Developing Solar PV Farm) = 4.317 * 0.3 + 4.905* 0.4 + 5.272 * 0.3 = 4.8387

EU (Expected Utility) = max(4.8148, 4.8387) = 4.8387

This shows that Strategy 3: Developing the Solar PV Farm has the highest expected utility, after comparing the expected utilities of other strategies.

EMV (Expected Monetary Value): 135

CE (Certainty Equivalent):

$$\ln(x) = 4.8387$$

$$e^{4.8387} = x$$

$$x = CE = 126.305$$

RP (Risk Premium): RP = EMV − CE = 135 − 126.305 = 8.695

A positive risk premium suggests that the decision-maker is willing to sacrifice part of the expected monetary value to avoid the uncertainty associated with risk. This behavior is illustrated by the concave shape of the utility function, which indicates a strong preference for certainty, even if it means accepting lower returns.

5.2. RISK-NEUTRAL DECISION MAKER

To analyse the choices of a risk-neutral decision maker, we employ the linear utility function

$$U(x) = 5x + 7$$

a positive affine transformation that leaves all monetary orderings intact. Because the slope is constant, every extra dollar delivers exactly the same amount of satisfaction. This property implies zero risk aversion: the decision maker is indifferent to variability and bases decisions solely on expected monetary value (EMV).

Table 5.2. Utilities of the consequences based on the utility function of U(x) = 5x+7

x	U(x) = 5x+7
79	402
139	702
199	1002
61	312
121	612
181	912
65	332
125	632
185	932
75	382
135	682
195	982

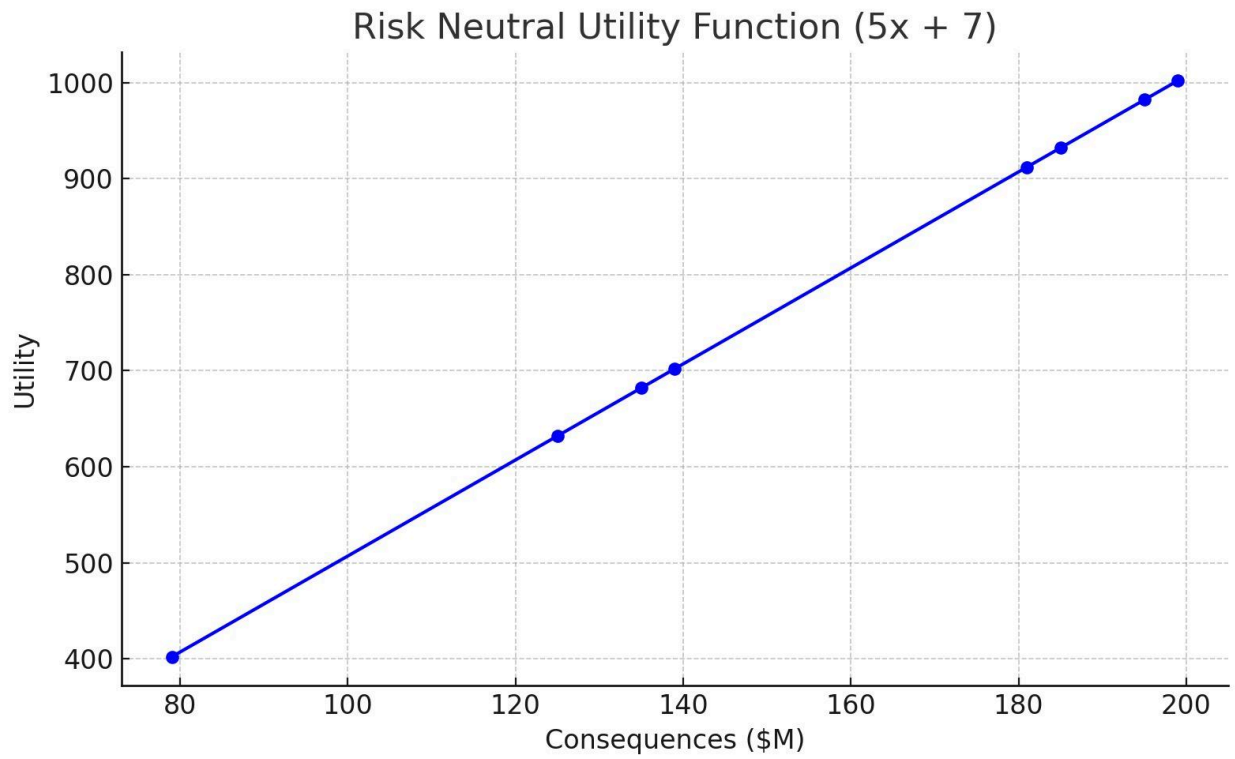


Figure 5.2. Plot of the utilities of the consequences based on the utility function U(x) = 5x+7

Figure 5.2 would display the straight-line relationship between wealth and utility. The line's unchanging slope illustrates constant marginal utility; therefore, two prospects with equal EMVs also have equal expected utilities (EUs). In this setting, the entire decision problem collapses to selecting the strategy with the highest EMV.

$$\rho = \frac{-U''(x)}{U'(x)}$$

U'(x) = 5

U''(x) = 0

Coefficient of Risk Aversion = 0 / 5 = 0 (CARA)

U(3.8 Interest Rate) = 402 * 0.2 + 702 * 0.5 + 1002 * 0.3 = 732

U(4.2 Interest Rate) = 312 * 0.2 + 612* 0.5 + 912 * 0.3 = 642

U(Issuing Green Bonds) = 732 * 0.2 + 642 * 0.8 = 660

U(Obtaining Bank Loan) = 332 * 0.2 + 632 * 0.5 + 932 * 0.3 = 662

U(Building Natural Gas Plant) = max(660, 662) = 662

U(Developing Solar PV Farm) = 382 * 0.3 + 682* 0.4 + 982 * 0.3 = 682

EU (Expected Utility) = max(662, 682) = 682

This shows that Strategy 3: Developing the Solar PV Farm has the highest expected utility, after comparing the expected utilities of other strategies.

EMV (Expected Monetary Value): 135

CE (Certainty Equivalent):

$$5x+7 = 682$$

$$5x = 675$$

$$x = CE = 135$$

RP (Risk Premium): $RP = EMV - CE = 135 - 135 = 0$

A risk premium of zero indicates that the decision-maker is risk-neutral. This means they are indifferent between a certain outcome and a risky option with the same expected monetary value. In other words, they do not require extra compensation to bear risk, nor are they willing to sacrifice potential return to avoid it. Their decisions are based solely on expected value, without regard for variability or uncertainty.

5.3. RISK-SEEKING DECISION MAKER

To capture the preferences of a risk-seeking investor we use the convex utility function

$$U(x) = \frac{1}{2}x^2$$

under which larger monetary outcomes deliver *disproportionately* higher satisfaction. Because marginal utility increases with wealth, the decision maker values upside potential more than downside loss and is willing to tolerate—or even seek out—high variability.

Table 5.3. Utilities of the consequences based on the utility function of $U(x) = ½*x^2$

x	U(x) = 1/2x^2
79	3120,5
139	9660,5
199	19800,5
61	1860,5
121	7320,5
181	16380,5
65	2112,5
125	7812,5
185	17112,5
75	2812,5
135	9112,5
195	19012,5

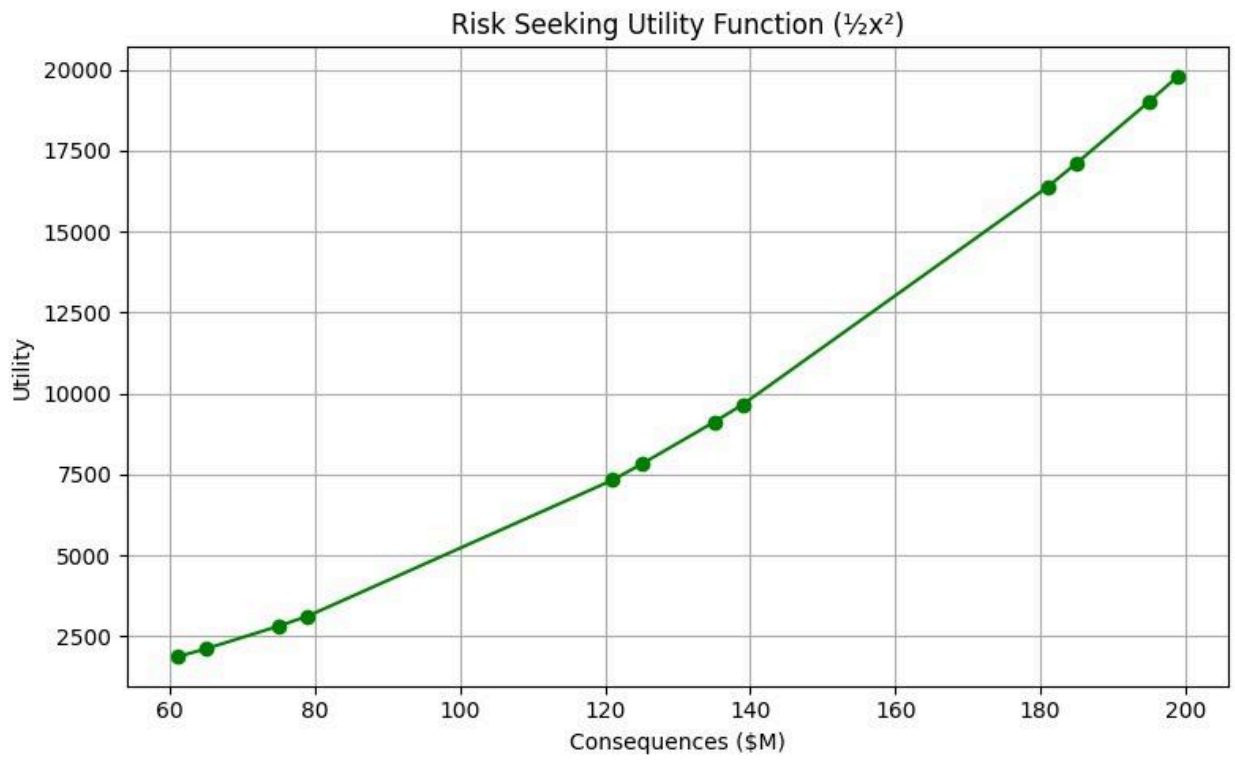


*Figure 5.3. Plot of the utilities of the consequences based on the utility function U(x) = ½*x²*

Figure 5.3 reveals that U(x) = ½*x² is convex, so marginal utility increases—each extra dollar delivers more satisfaction than the previous one. A decision-maker with this utility embraces volatility: the promise of a very large payoff outweighs concern for possible losses. Consequently, a risk-seeking investor may prefer a high-variance project with a lower EMV over a safer alternative if it offers even a small chance of exceptionally large gains.

$$\rho = \frac{-U''(x)}{U'(x)}$$

$$U'(x) = \frac{1}{x}$$

$$U''(x) = -\frac{1}{x^2}$$

Coefficient of Risk Aversion = - (-1/x²) / (1/x) = 1/x (DARA)

U(3.8 Interest Rate) = 3120.5 * 0.2 + 9660.5 * 0.5 + 19800.5 * 0.3 = 11394.5

U(4.2 Interest Rate) = 1860.5 * 0.2 + 7320.5* 0.5 + 163.80 * 0.3 = 4081.49

U(Issuing Green Bonds) = 11394.5 * 0.2 + 4081.49 * 0.8 = 5544.092

U(Obtaining Bank Loan) = 2112.5 * 0.2 + 7812.5 * 0.5 + 17112.5 * 0.3 = 9462.5

U(Building Natural Gas Plant) = max(5544.092, 9462.5) = 9462.5

U(Developing Solar PV Farm) = 2812.5 * 0.3 + 9112.5* 0.4 + 19012.5 * 0.3 = 10192.5

EU (Expected Utility) = max(9462.5, 10192.5) = 10192.5

This shows that Strategy 3: Developing the Solar PV Farm has the highest expected utility, after comparing the expected utilities of other strategies.

EMV (Expected Monetary Value): 135

CE (Certainty Equivalent):

$\frac{1}{2}*x^2 = 10192.5$

$x^2 = 20385$

$x = CE = 142.776$

RP (Risk Premium): RP = EMV − CE = 135 − 142.776 = -7.776

A negative risk premium indicates that the decision-maker is willing to accept a lower average monetary value in exchange for the chance to pursue a risky option with potentially high rewards. This reflects a risk-seeking attitude, where the possibility of a large gain outweighs the appeal of a more certain but smaller outcome.

Therefore, it's justified to conclude: All decision-makers choose Strategy 3, as it maximizes their expected utility.

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